

Wireshark Network Analysis

Conversations Tab — Full Study Notes

A complete record of the capture data, analysis, and lessons covered in this session.

Section 1: Raw Capture Data

Ethernet Tab (2 Conversations)

Address A	Address B	Packets	Bytes	Pkts A→B	Bytes A→B	Pkts B→A	Bytes B→A	Rel Start	Duration	Bits/s A→B	Bits/s B→A
5d:b2:bf:7a:7b	50:5a:65:4f:14:e9	1,153	1 MB	756	913 kB	397	125 kB	0.000000	9.407756	775 kbps	106 kbps
5a:65:4f:14:e9	ff:ff:ff:ff:ff:ff	1	86 B	1	86 B	0	0 B	0.051689	0.000000	—	—

IPv4 Tab (5 Conversations)

Address A	Address B	Packets	Bytes	Pkts A→B	Bytes A→B	Pkts B→A	Bytes B→A	Rel Start	Duration	Bits/s A→B	Bits/s B→A
11.241.224	10.238.64.91	2	108 B	1	54 B	1	54 B	2.422407	0.000031	—	—
10.238.64.91	10.238.64.97	18	2 kB	9	768 B	9	2 kB	1.509528	7.326159	838 b/s	1871 b/s
10.238.64.91	10.238.64.255	1	86 B	1	86 B	0	0 B	0.051689	0.000000	—	—
10.238.64.91	132.196.74.212	180	203 kB	55	58 kB	125	145 kB	1.633939	6.319882	73 kbps	183 kbps
10.238.64.91	142.250.77.110	21	10 kB	9	5 kB	12	5 kB	8.836572	0.313137	121 kbps	122 kbps

IPv6 Tab (13 Conversations)

Address A	Address B	Packets	Bytes	Pkts A→B	Bytes A→B	Pkts B→A	Bytes B→A	Rel Start	Duration
2401:4900:::6bed	2404:6800:::2003 (82e)	2	161 B	1	75 B	1	86 B	4.623555	0.060071
2401:4900:::6bed	2404:6800:::200a (836)	4	569 B	2	185 B	2	384 B	2.528651	5.021409
2401:4900:::6bed	2404:6800:::2003 (838)	2	161 B	1	75 B	1	86 B	1.193158	0.042035
2401:4900:::6bed	2600:1901:1:388::	44	20 kB	21	11 kB	23	8 kB	1.594921	5.473222
2401:4900:::6bed	2600:1901:1:7c5::	7	2 kB	4	2 kB	3	450 B	5.224510	0.369762
2401:4900:::6bed	2600:1901:1:d18::	4	379 B	2	191 B	2	188 B	1.929551	0.283798
2401:4900:::6bed	2a04:4e42:8e::684	3	222 B	3	222 B	0	0 B	7.881746	1.395917
2401:4900:::6bed	2a04:4e42:92::684	1	74 B	1	74 B	0	0 B	4.512654	0.000000
2404:6800:::200e (834)	2401:4900:::6bed	2	221 B	1	147 B	1	74 B	8.562349	0.041091
2600:140f:::3791	2401:4900:::6bed	834	791 kB	559	747 kB	275	44 kB	0.000000	9.407756
2600:14e1:1:c6d::	2401:4900:::6bed	2	160 B	1	86 B	1	74 B	2.946259	2.871524
2620:1ec:bdf::72	2401:4900:::6bed	7	581 B	4	359 B	3	222 B	6.999898	0.049626
2a03:2880:::167	2401:4900:::6bed	18	7 kB	11	4 kB	7	3 kB	0.071599	1.566284

TCP Tab (18 Conversations)

Address A	Port A	Address B	Port B	Packets	Bytes	Pkts A→B	Bytes A→B	Pkts B→A	Bytes B→A	Rel Start
3.111.241.224	443	10.238.64.91	49372	2	108 B	1	54 B	1	54 B	2.422407

Address A	Port A	Address B	Port B	Packets	Bytes	Pkts A→B	Bytes A→B	Pkts B→A	Bytes B→A	Rel Start
10.238.64.91	65032	132.196.74.212	443	180	203 kB	55	58 kB	125	145 kB	1.633939
2401:.....6bed	62118	2404:.....2003 (82e)	443	2	161 B	1	75 B	1	86 B	4.623555
2401:.....6bed	56831	2404:.....2003 (838)	443	2	161 B	1	75 B	1	86 B	1.193158
2401:.....6bed	60084	2600:14e1:1:c6d::	80	1	74 B	1	74 B	0	0 B	5.817783
2401:.....6bed	49640	2600:1901:1:388::	443	13	1 kB	6	712 B	7	729 B	6.749434
2401:.....6bed	58151	2600:1901:1:d18::	443	4	379 B	2	191 B	2	188 B	1.929551
2401:.....6bed	54042	2a04:4e42:8e::684	80	1	74 B	1	74 B	0	0 B	8.493021
2401:.....6bed	54059	2a04:4e42:8e::684	80	1	74 B	1	74 B	0	0 B	9.277663
2401:.....6bed	54064	2a04:4e42:8e::684	80	1	74 B	1	74 B	0	0 B	7.881746
2401:.....6bed	54063	2a04:4e42:92::684	80	1	74 B	1	74 B	0	0 B	4.512654
2404:.....200e (834)	443	2401:.....6bed	60063	2	221 B	1	147 B	1	74 B	8.562349
2600:140f:.....3791	80	2401:.....6bed	63295	834	791 kB	559	747 kB	275	44 kB	0.000000
2600:14e1:1:c6d::	80	2401:.....6bed	63300	2	86 B	1	86 B	0	0 B	2.946259
2620:1ec:bdf::72	443	2401:.....6bed	60094	7	581 B	4	359 B	3	222 B	6.999898
2a03:2880:.....167	443	2401:.....6bed	63292	3	330 B	1	330 B	0	0 B	9.259598
2a03:2880:.....167	443	2401:.....6bed	64907	1	330 B	1	330 B	0	0 B	1.296717
2a03:2880:.....167	5222	2401:.....6bed	49796	3	1 kB	2	872 B	1	139 B	0.071599

UDP Tab (16 Conversations)

Address A	Port A	Address B	Port B	Packets	Bytes	Pkts A→B	Bytes A→B	Pkts B→A	Bytes B→A	Rel Start
10.238.64.91	50481	10.238.64.97	53	2	247 B	1	85 B	1	162 B	6.710868
10.238.64.91	53164	10.238.64.97	53	2	452 B	1	86 B	1	366 B	8.778931
10.238.64.91	53180	10.238.64.97	53	2	298 B	1	85 B	1	213 B	1.509528
10.238.64.91	53590	10.238.64.97	53	2	235 B	1	85 B	1	150 B	6.710964
10.238.64.91	55456	10.238.64.97	53	2	236 B	1	85 B	1	151 B	1.510094
10.238.64.91	56572	10.238.64.97	53	2	298 B	1	85 B	1	213 B	6.710741
10.238.64.91	57146	10.238.64.97	53	2	246 B	1	85 B	1	160 B	8.778562
10.238.64.91	58261	10.238.64.97	53	2	246 B	1	86 B	1	160 B	8.778770
10.238.64.91	58269	10.238.64.97	53	2	224 B	1	85 B	1	139 B	1.510318
10.238.64.91	57621	10.238.64.255	57621	1	86 B	1	86 B	0	0 B	0.051689
10.238.64.91	62019	142.250.77.110	443	21	10 kB	9	5 kB	12	5 kB	8.836572
2401:.....6bed	61963	2404:.....200a (836)	443	4	569 B	2	185 B	2	384 B	2.528651
2401:.....6bed	53108	2600:1901:1:388::	443	11	8 kB	5	4 kB	6	4 kB	6.748939
2401:.....6bed	62891	2600:1901:1:388::	443	20	10 kB	10	6 kB	10	4 kB	1.594921

Address A	Port A	Address B	Port B	Packets	Bytes	Pkts A→B	Bytes A→B	Pkts B→A	Bytes B→A	Rel Start
2401:::6bed	56055	2600:1901:1:7c5::	443	7	2 kB	4	2 kB	3	450 B	5.224510
2401:::6bed	49627	2a03:2880:::167	443	13	5 kB	6	3 kB	7	2 kB	1.231239

Section 2: Capture Analysis

Identifying Your Device

Based on multiple clues across all tabs, the device generating the capture is almost certainly:

Layer	Identifier
IPv4	10.238.64.91
IPv6	2401:4900:91c8:7c80:dc7d:bcf0:58eb:6bed
MAC	50:5a:65:4f:14:e9 (AzureWave Wi-Fi adapter)

Everything revolves around these addresses. On Wi-Fi, almost all Layer 2 (Ethernet) traffic appears as Laptop ↔ Router, even when the actual destination is Google or another external server.

IPv4 Conversation Breakdown

10.238.64.91 → 10.238.64.255 — 1 packet, 86 bytes

Broadcast packet. Typical of ARP or service discovery. Only one packet suggests a one-off ARP request. Nothing suspicious.

10.238.64.91 ↔ 10.238.64.97 — 18 packets, 2 kB, 7.3 seconds

Local network communication. Equal packet count (9 each way) is consistent with request/response. Confirmed as DNS traffic on port 53 by the UDP tab.

3.111.241.224 ↔ 10.238.64.91 — 2 packets, 108 bytes, 0.000031 s

Extremely short exchange — one packet each way. Likely a probe, ICMP ping, or failed connection attempt. Very brief, not a sustained session.

10.238.64.91 ↔ 132.196.74.212 — 180 packets, 203 kB — 58 kB sent, 145 kB received

Largest IPv4 session. Classic client/server download pattern: small request, large response. Consistent with web browsing over HTTPS (port 443 confirmed in TCP tab).

10.238.64.91 ↔ 142.250.77.110 — 21 packets, 10 kB, 0.3 seconds

IP range 142.250.x.x is commonly associated with Google infrastructure. Traffic runs over UDP port 443, which is the fingerprint of QUIC / HTTP3. Normal if a browser was open.

Notable TCP Findings

The TCP tab reveals which ports and services were used:

- Port 443 (HTTPS) dominates — multiple servers contacted over TLS.
- Port 80 (HTTP) appears for some connections, likely redirect or CDN probes.
- Port 5222 from 2a03:2880:f334:120:face:b00c:0:167 — the 2a03:2880 prefix is associated with Meta Platforms. Port 5222 is used by messaging protocols (XMPP-based). This suggests Facebook, Messenger, or a Meta service was active.

- 2600:140f:d000::17c9:3791 on port 80 — the dominant stream: 834 packets, 791 kB, with 747 kB received. Classic download pattern.

DNS Activity (UDP Tab)

Nine separate DNS conversations to 10.238.64.97:53. Each follows the same two-packet pattern: one query out, one response back. This is completely normal — every website visit triggers at least one DNS lookup. The local DNS server at 10.238.64.97 resolved all queries.

QUIC / HTTP3 (UDP Port 443)

Traffic to 142.250.77.110 (Google) runs over UDP port 443 — not TCP. This is QUIC, the protocol underlying HTTP/3. Modern browsers use QUIC for Google services by default. An important Wireshark recognition skill: UDP + port 443 is not the same as TCP + port 443.

Section 3: What to Focus On for Ethical Hacking

Six Key Questions for Any Conversations Tab

Question	What to look for
1. Who is talking to whom?	Identify the two endpoints. Is it local-to-local or local-to-internet?
2. Which side is the client and which is the server?	High ephemeral port (e.g. 65032) = client. Well-known port (e.g. 443) = server.
3. Which protocol/service is in use?	53=DNS, 80=HTTP, 443=HTTPS/QUIC, 22=SSH, 25=SMTP, 5222=Messaging.
4. Who sent more data?	Large upload could indicate exfiltration. Large download is typical web browsing.
5. Which conversations are most active?	Prioritise by packet count and byte volume. Investigate the biggest streams first.
6. Can I isolate this conversation?	Select row → Apply as Filter → Follow TCP Stream. The most practical skill in Wireshark.

Wireshark Knowledge Priority for Ethical Hacking

Priority	Topic
Must Know	TCP vs UDP
Must Know	Common ports (53, 80, 443, 22, 25, 5222)
Must Know	DNS — how queries and responses work
Must Know	HTTP / HTTPS
Must Know	TCP 3-way handshake
Must Know	Follow TCP Stream
Must Know	Conversations tab
Must Know	Endpoints tab
Useful	Protocol Hierarchy
Useful	ARP details
Useful	DHCP details
Useful	QUIC / HTTP3 (UDP 443)
Lower Priority	Resolved Addresses
Lower Priority	Most graph/statistics windows

Conversations Tab Priority Order

Rank	Tab	Why
1	TCP Conversations	Shows ports, services, data volume, client/server roles
2	UDP Conversations	DNS, QUIC/HTTP3, other UDP services
3	IPv4 / IPv6 Conversations	IP-level view, useful for spotting external destinations
4	Ethernet Conversations	Useful mainly for ARP spoofing, Wi-Fi, and Layer 2 issues

Section 4: Identifying Suspicious Traffic

The Conversations tab is a hypothesis generator, not a proof engine. Use it to decide which packets deserve closer inspection.

Unusual Ports

Normal: 53, 80, 443, 22. Anything else deserves a second look.

Example: 192.168.1.5 → 203.0.113.10:4444

Port Scanning

One host communicating with many ports on another host.

Example: 192.168.1.5 → 192.168.1.1: ports 21, 22, 80, 443, 445

Beaconing

Repeated connections to the same external host at regular intervals.

Example: 192.168.1.5 → 198.51.100.50 every 30 seconds

Large Data Imbalance

Very large upload (send >> receive) may indicate data exfiltration.

Example: Sent: 500 MB | Received: 20 KB

Unknown External Destinations

Unrecognised IPs appearing frequently.

Example: 192.168.1.5 → 203.0.113.44 (unknown, repeated)

Sudden Spike in Conversations

Device communicating with hundreds of hosts when it normally uses 5-10.

Example: Possible scan, malware, or misconfiguration

Protocol Mismatch

A printer creating hundreds of external connections. An IoT device transferring large amounts of data.

Example: Context-dependent — know your environment

Assessment of This Capture

Item	Finding
DNS traffic	Normal — port 53, request/response pattern
HTTPS (TCP 443)	Normal — standard web browsing
QUIC (UDP 443)	Normal — Google services via HTTP/3
Meta/Facebook traffic	Normal — port 443 and 5222 for messaging
Port scanning	Not observed
Beaconing	Not observed

Item	Finding
Unusual ports	None identified
Suspiciously large transfers	None — largest is consistent with downloading web content
Overall verdict	Consistent with ordinary web browsing and application use

Section 5: Analyst Reasoning Principles

Clues vs Proof

A key theme throughout this session: statistics windows and timestamp comparisons generate hypotheses. Packet details provide evidence. Always distinguish between the two.

Source of information	Role
Conversations / Statistics tabs	Generate hypotheses — point you toward interesting traffic
Rel Start timestamps	Context — suggest ordering, not causation
Packet contents (detail pane)	Evidence — the only source of proof
Protocol field in packet	Proof that a packet is ARP, DNS, TLS, etc.

Identifying Your Device — How It Was Done

No single clue proved which MAC/IP belonged to the capture machine. The conclusion came from combining multiple weak signals:

- MAC 50:5a:65:4f:14:e9 resolved to AzureWave (common laptop Wi-Fi adapter vendor).
- IPv4 address 10.238.64.91 appeared in every conversation as the initiating side (making DNS queries, opening HTTPS connections).
- The broadcast packet originated from 50:5a:65:4f:14:e9, suggesting it is an end device.
- On Wi-Fi, nearly all Ethernet-level traffic is between laptop and access point — confirmed by only two MAC addresses in the Ethernet tab.

To prove it definitively, you would need: an ARP packet showing '10.238.64.91 is-at 50:5a:65:4f:14:e9', or a DHCP packet, or the Endpoints tab mapping IPs to MACs.

The Broadcast Packet — Lesson in Weak Evidence

The broadcast packet (50:5a:65:4f:14:e9 → ff:ff:ff:ff:ff:ff) was initially suggested as evidence of ARP. The correct reading:

- Any device on the LAN can send a broadcast — it is not exclusive to the capture machine.
- The packet size (86 bytes) is consistent with ARP, but does not prove it.
- The Rel Start (0.051 s) precedes the DNS conversation (1.509 s), but timing alone does not establish causation.
- The broadcast packet could be completely unrelated to the DNS traffic — the capture may have started mid-session.
- Only the packet detail pane showing 'Protocol: ARP' would constitute proof.

The Rel Start Field

Rel Start records when the first packet of a conversation appeared in the capture file — not the absolute time, and not when the underlying session began. The session may have started before the capture. Use Rel Start as context, not as causal evidence.

Analogy: knowing Person A entered a building before Person B tells you nothing about whether they know each other or whether their actions are connected.

Analyst Workflow After Spotting Something Interesting

Step	Action
1	Identify the conversation in the Conversations tab (high packet count, unusual port, unknown IP)
2	Apply it as a display filter (right-click → Apply as Filter)
3	Examine the individual packets in the packet list
4	Follow the TCP (or UDP) stream for the full payload
5	Review protocol details — confirm what the traffic actually is
6	Determine whether the activity is legitimate or suspicious